

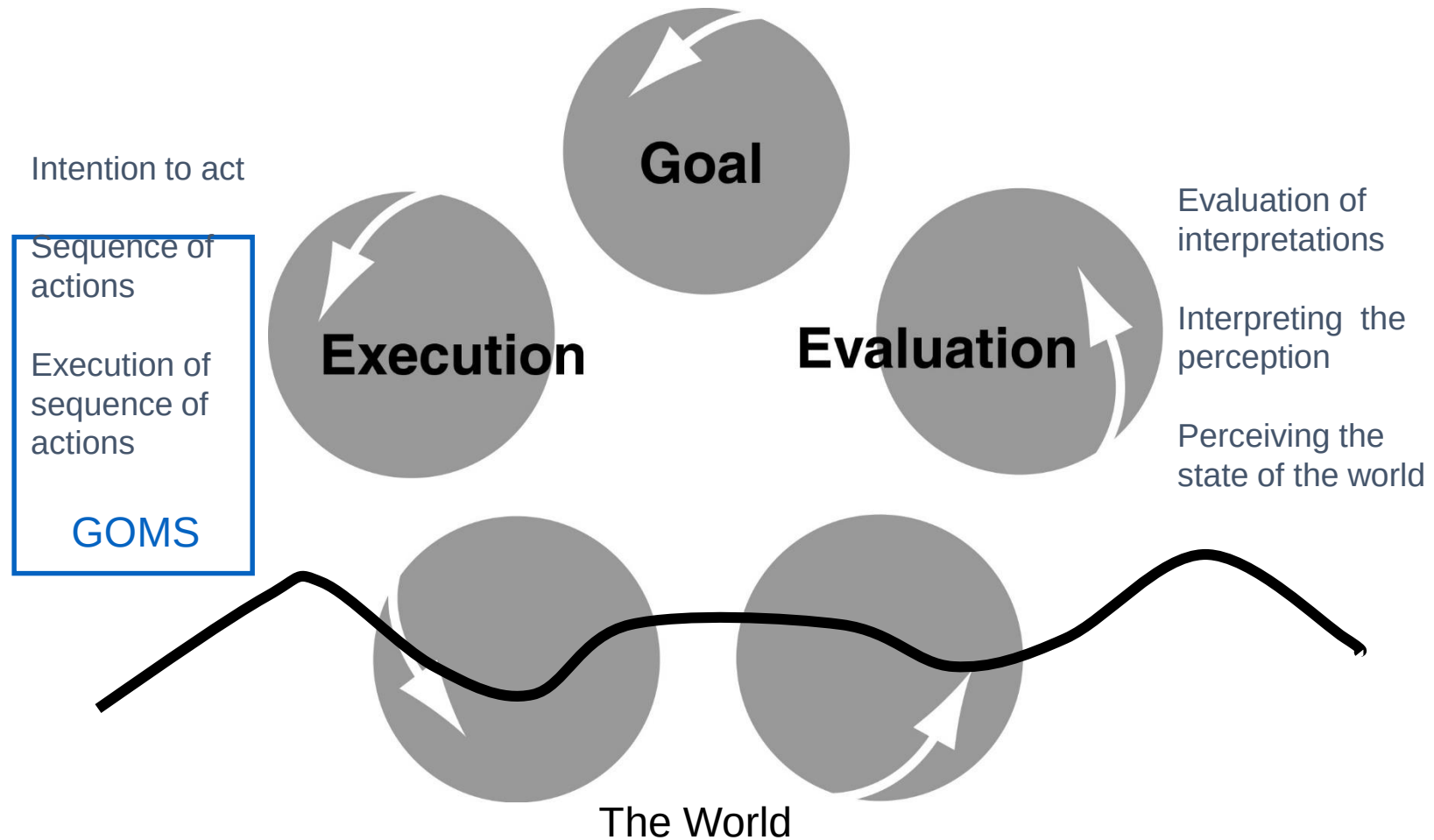
KLM

Dr. Ann Nosseir

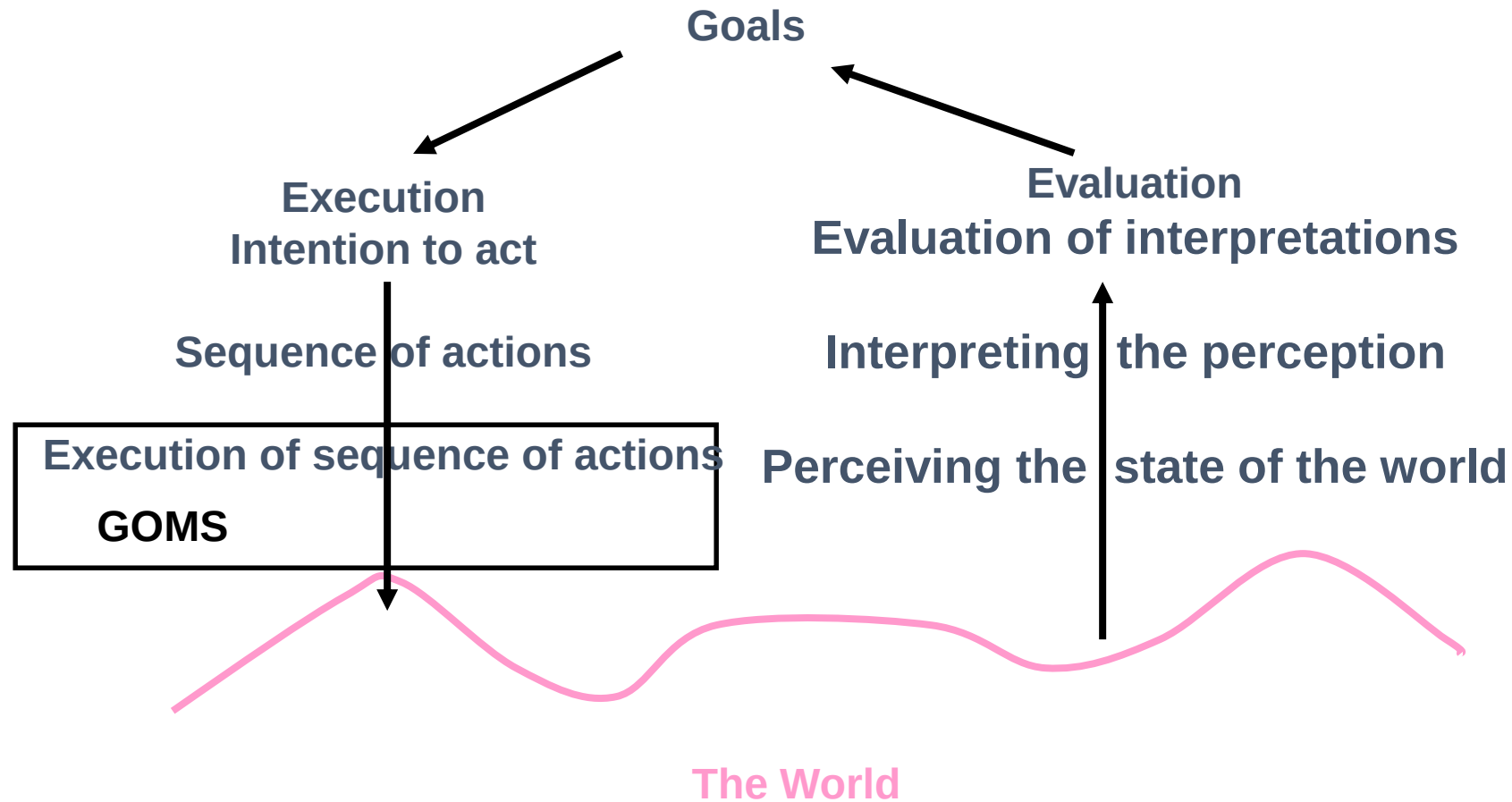
2. GOMS

- **G**oals, **O**perators, **M**ethods, **S**election Rules
 - Developed by Card, Moran and Newell
- Probably the most widely known and used technique in this family

Norman's Human Action Cycle (1988)



Norman's Action Cycle



Norman's Seven-Stage Model of Activity

- **Gulf of execution:** refers to the problem of how an individual translates intentions into action
 - A gap between the psychological language (or mental model) of the user's goals and the very physical action-object language of the controls of the VCR via which it is operated.



User

Hit the record button



VCR

1. Hit the record button.
2. Specify time of recording via the controls X, Y, and Z.
3. Select channel via the channel-up-down control.
4. Press the OK button.

GOMS analysis

- Engineering model of user interaction--task analysis
 - Goals - user's intentions (tasks)
 - e.g. delete a file, edit text, assist a customer
 - Operators - actions to complete task
 - cognitive, perceptual & motor (MHP)
 - low-level (e.g. move the mouse to menu)
 - Methods - sequences of actions (operators)
 - based on error-free expert
 - may be multiple methods for accomplishing same goal
 - e.g. shortcut key or menu selection
 - Selections - rules for choosing appropriate method
 - method predicted based on context

CMN-GOMS

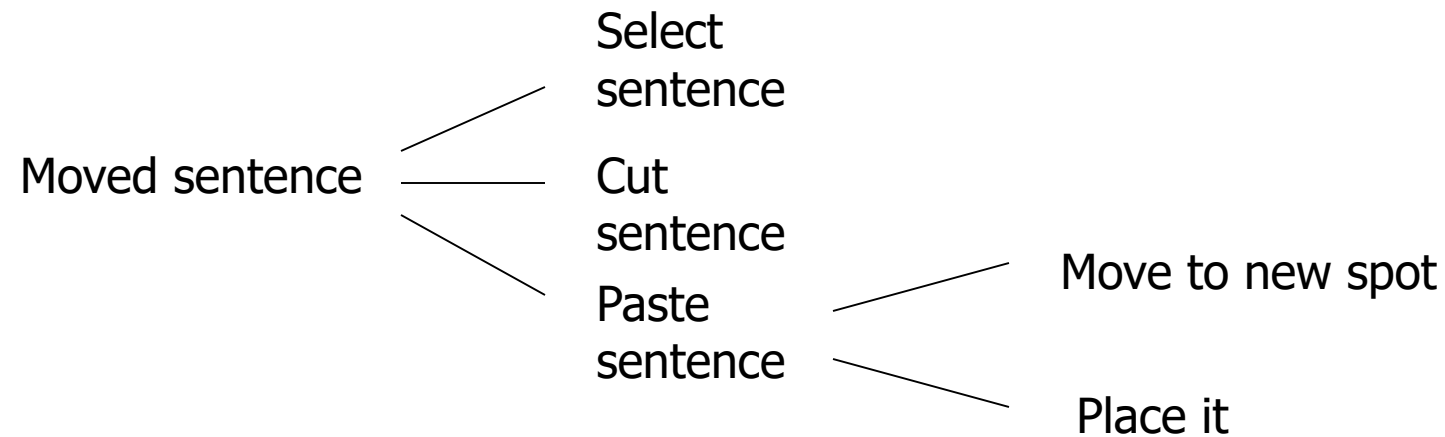
- Engineering model of user interaction (continued)
 - task analysis (“how to” knowledge)
 - Methods - sequences of actions (operators)
 - based on error-free expert
 - may be multiple methods for accomplishing same goal
 - e.g., shortcut key or menu selection
 - Selections - rules for choosing appropriate method
 - method predicted based on context
 - explicit task structure
 - hierarchy of goals & sub-goals

Assumptions

- “Expert” is performing UI operations
- Interacting with system is problem solving
- Decompose into subproblems
- Determine goals to attack problem
- Know sequence of operations used to achieve the goals
- Timing values for each operation

Goal

- End state trying to achieve
- Then decompose into subgoals

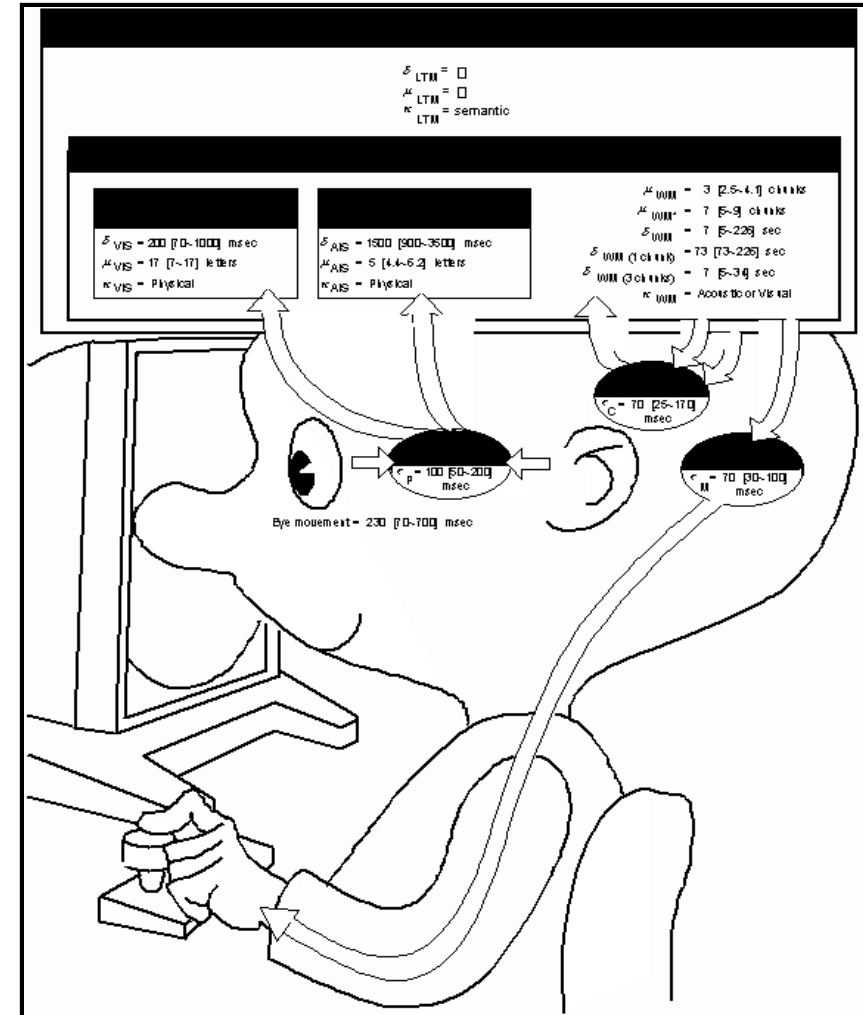


Operators

- Basic actions available for performing a task (lowest level actions)
- Examples: move mouse pointer, drag, press key, read dialog box, ...

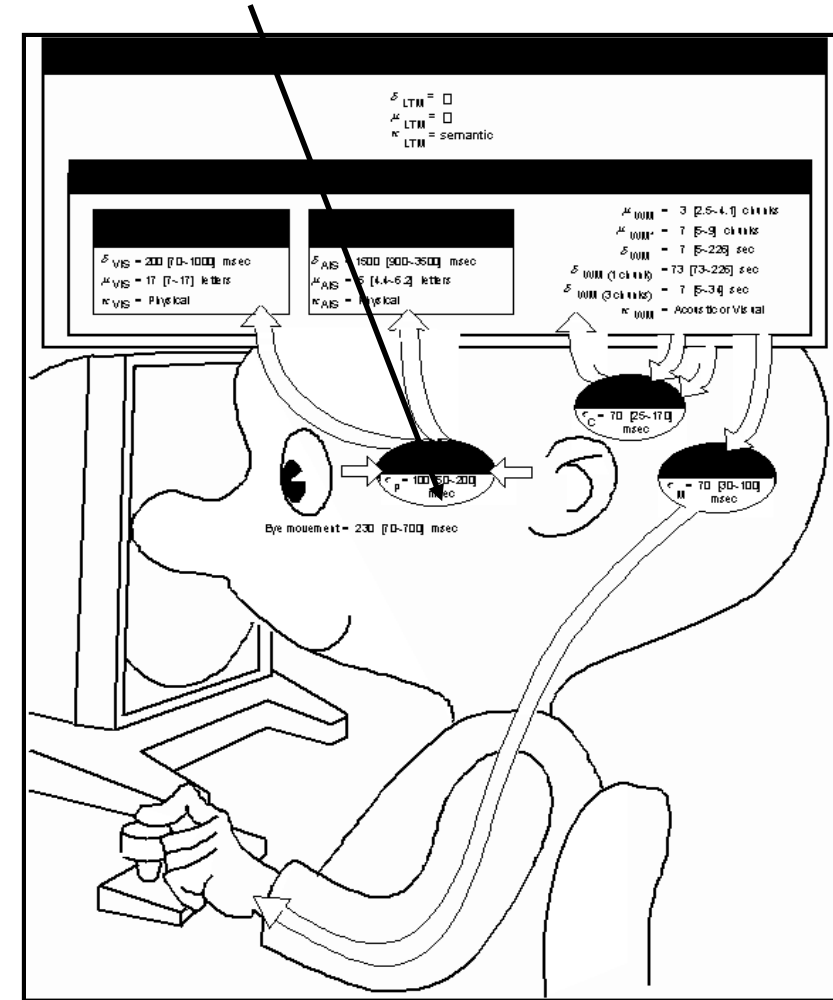
Model Human Processor (MHP) Card, Moran & Newell (1983)

- Most influential model of user interaction
- 3 interacting subsystems:-
cognitive, perceptual & motor
 - each with processor & memory
 - serial & parallel processing



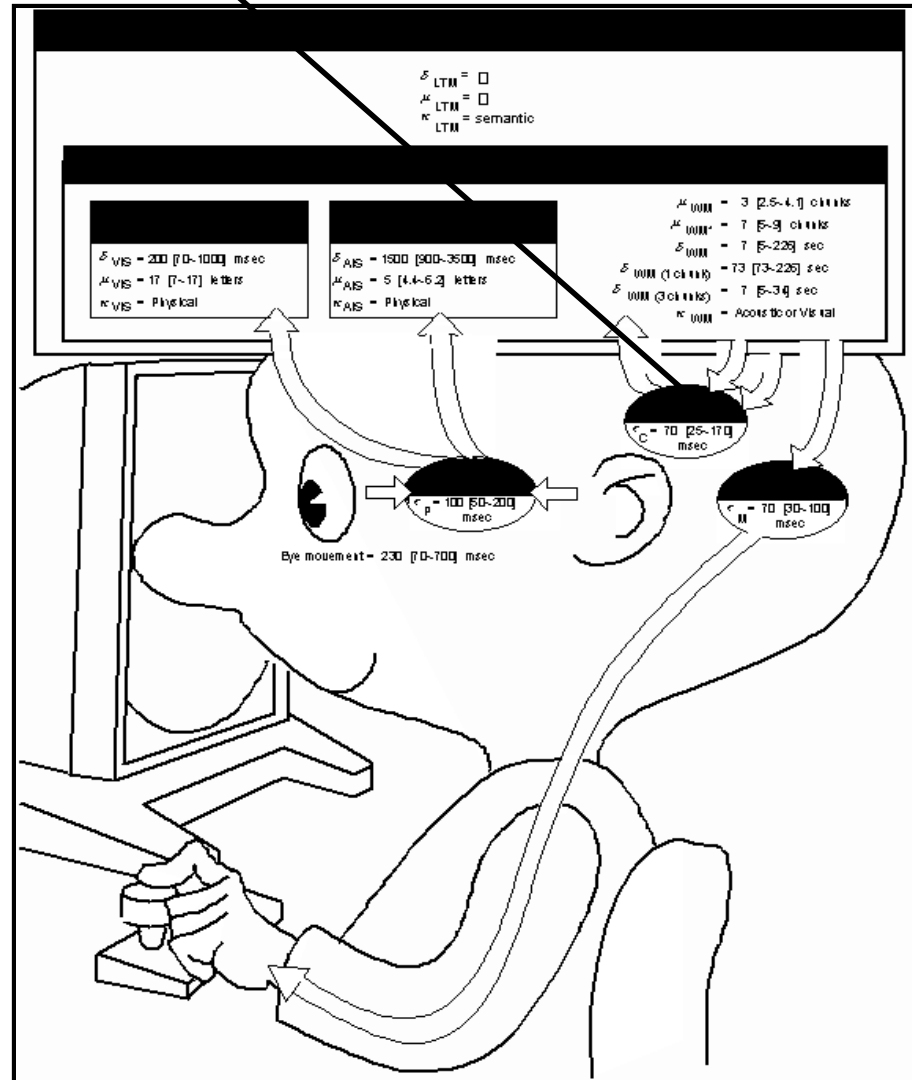
Perceptual Processor

- Sensory input (audio & visual)
- Code info symbolically
- Output into audio & visual image storage (WM buffers)



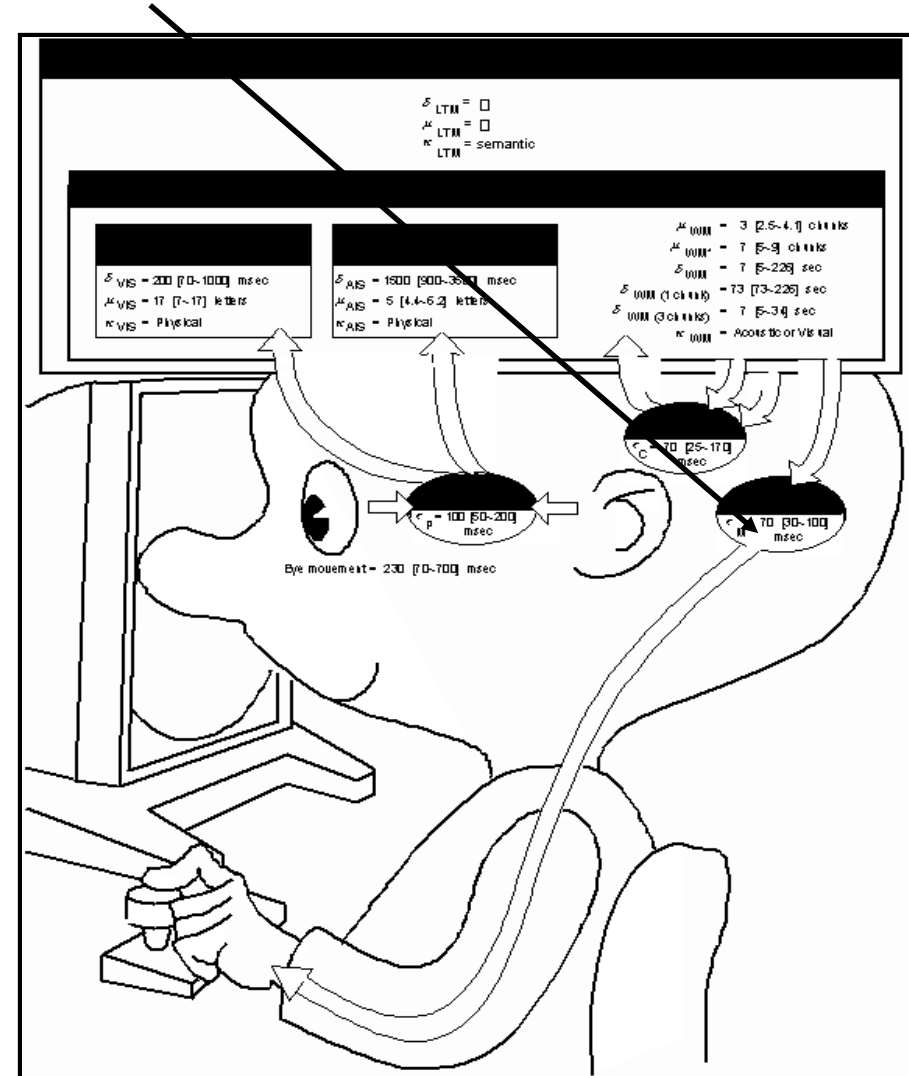
Cognitive Processor

- Input from sensory buffers
- Access LTM to determine response- previously stored info
- Output response into WM



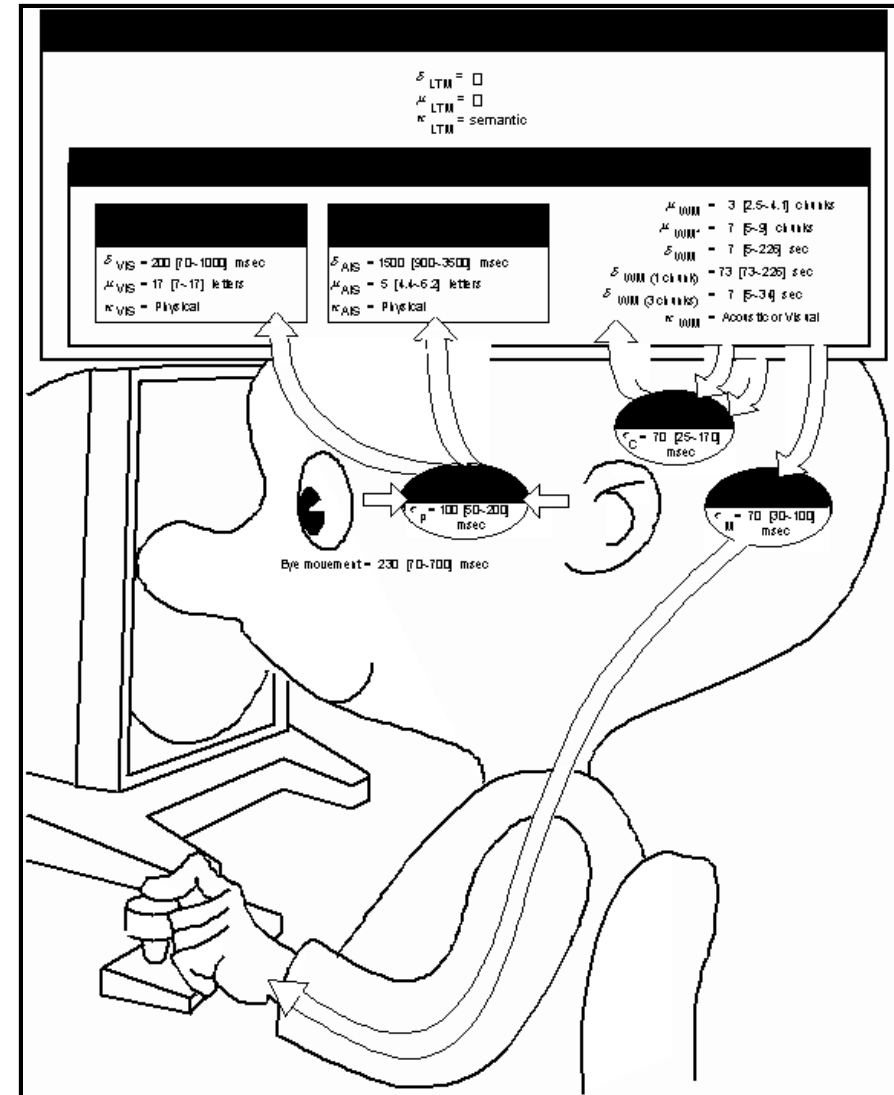
Motor Processor

- Input response from WM
- Carry out response



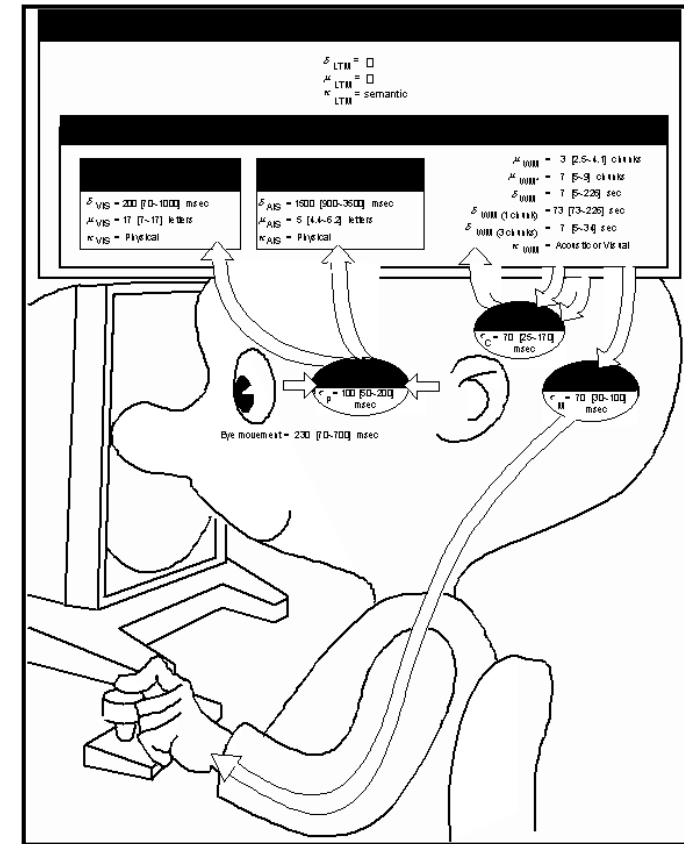
MHP Subsystem Interactions

- Input/output
- Processing
 - serial action
 - pressing key in response to light
 - parallel perception
 - driving, reading signs & hearing



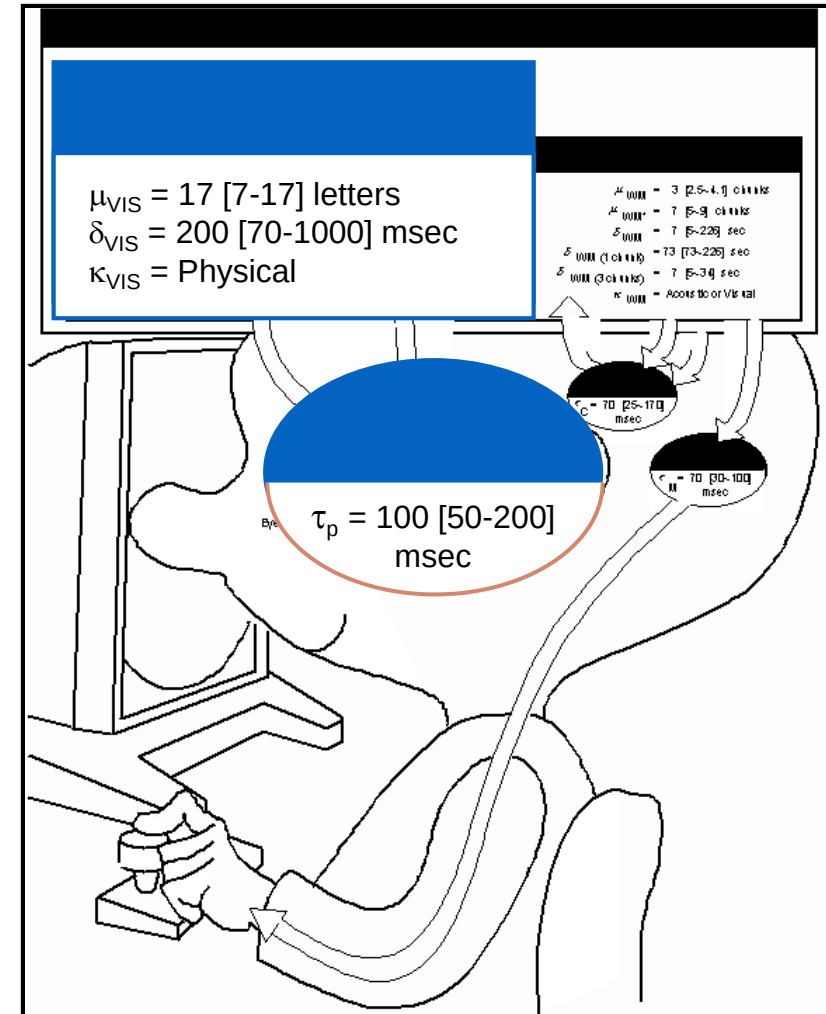
MHP Parameters

- Based on empirical data
 - word processing in the '70s
- Processors have
 - cycle time (τ)
- Memories have
 - storage capacity (μ)
 - decay time of an item (δ)
 - info code type (κ)
 - physical, acoustic, visual & semantic



Perceptual Subsystem Parameters

- Processor
 - cycle time (τ) = 100 msec
- Visual Image Store
 - storage capacity (μ) = 17 letters
 - decay time of an item (δ) = 200 msec
 - info code type (κ) = physical
 - physical properties of visual stimulus
 - e.g., intensity, color, curvature, length
- Auditory Image Store
 - similar parameters

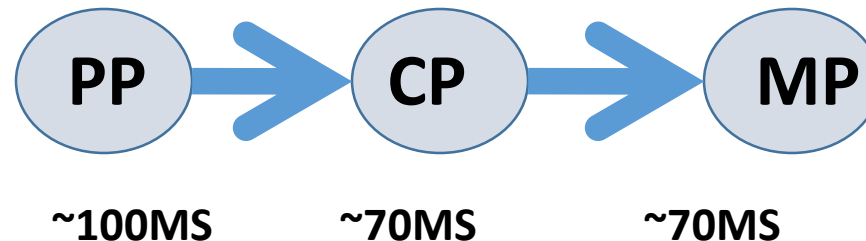


One Principle of Operation

- Power Law of Practice
 - task time on the nth trial follows a power law
 - $T_n = T_1 n^{-a}$, where $a = .4$
 - i.e., you get faster the more times you do it!
 - applies to skilled behavior (perceptual & motor)
 - does not apply to knowledge acquisition or quality

Predictive Evaluation

- Predictive evaluation uses an engineering model of human cognition to predict usability
- Model is abstract , quantitative, approximate , and estimated from user experiments



Keystroke-Level Model (KLM)

- **K**ey stroke
- **B**utton press or release with mouse
- **P**oint with mouse
- **D**raw Line with the mouse
- **H**ome hands between mouse and keyboard
- **M**entally prepare

KLM Analysis

- Encode a method as a sequence of physical operations
- Use heuristic rules to insert mental operators (M)
- Add up times for each operator to get total time for method

Estimated Operator Times

- **Keystroke** determinate by typing speed
(0.28 s average typist)
- **Button press or release**
(0.1 s)
- **Pointing** determined by Fitt's law
($T = a + b \log(d/s + 1) = a + b ID$)
- **Drawing** determined by steering law
(6.813/6.831)

Estimated Operator Times

- **Homing** estimated by measuring
0.36 s (between keyboard and mouse)
- **Metal preparation** estimated by measuring
1.35 s

Heuristic Rules for Adding M's

- Basic Idea :**M** before every chunk in the method that must be recalled or that involves decisions
 - Before each task
 - Deciding which way to do a task
 - Retrieving a chunk from memory (e.g., command name, file name, parameter value)
 - Finding something
 - Verifying entry or action result (e.g., before pressing ok)

Heuristics Rules for Placing Mental Operations:

- *Rule 0*: Initial insertion of candidate Ms:
 - Insert M before all Ks and Ps
- *Rule 1*: Deletion of anticipated Ms:
 - If P or K is fully anticipated by a preceding P or K then delete the middle M. For example moving the mouse to tap on the button; PMK => PK
- *Rule 2*: Deletion of Ms in cognitive units:
 - If a series of K represent a string then delete the middle Ks; for example type '1.2' is a cognitive unit; MKMKMK => MKKK

Heuristics Rules for Placing Mental Operations:

- *Rule 3:* Deletion of Ms before consecutive terminators:
 - If several delimiters are typed only keep the first M. For example if ')))' is terminator count only one M.
- *Rule 4:* Deletion of Ms that are terminators of commands:
 - If the terminator is a frequently used string, delete the M before the terminator; for example a command followed by return so the M before the K representing the return is deleted. But if the terminator delimits the arguments for a command string that vary keep the M. This represents checking that the command is correct.
- *Rule 5:* Deletion of overlapped Ms:
 - Do not count any portion of an M that overlaps with a command response. (This is the reason that a responsive interface only need respond in a second.)

Example

Move Sentence

1. Select sentence

Reach for mouse	H	0.40
Point to first word	P	1.10
Click button down	K	0.60
Drag to last word	P	1.20
Release	K	<u>0.60</u>
		3.90 secs

2. Cut sentence

Press, hold ^		Point to menu
Press and release 'x'	or	Press and hold mouse
Release ^		Move to "cut"
		Release

3. ...

Example

Deleting a Word

Shift-click selection

M
P (start of word)
BB (click)
M
P (end of word)
K (shift)
BB (click)
H (to keyboard)
M
K (Del)

Total : $3M+2P+4B+1K = 6.93 \text{ sec}$

Del key N times

M
P (start of word)
BB(click)
H
M
K(Del)
X n(length of word)

Total : $2M+P+2B+H+nK = 4.36+0.28n \text{ sec}$

KLM example

- **Homing:** Moving Hand to Keyboard or Mouse: 360ms
- **Clicking:** the Mouse: 230 ms
- **Pointing:** with the Mouse: 1100ms
- **Mental Operations:** (Deciding what to Do): 1350 ms

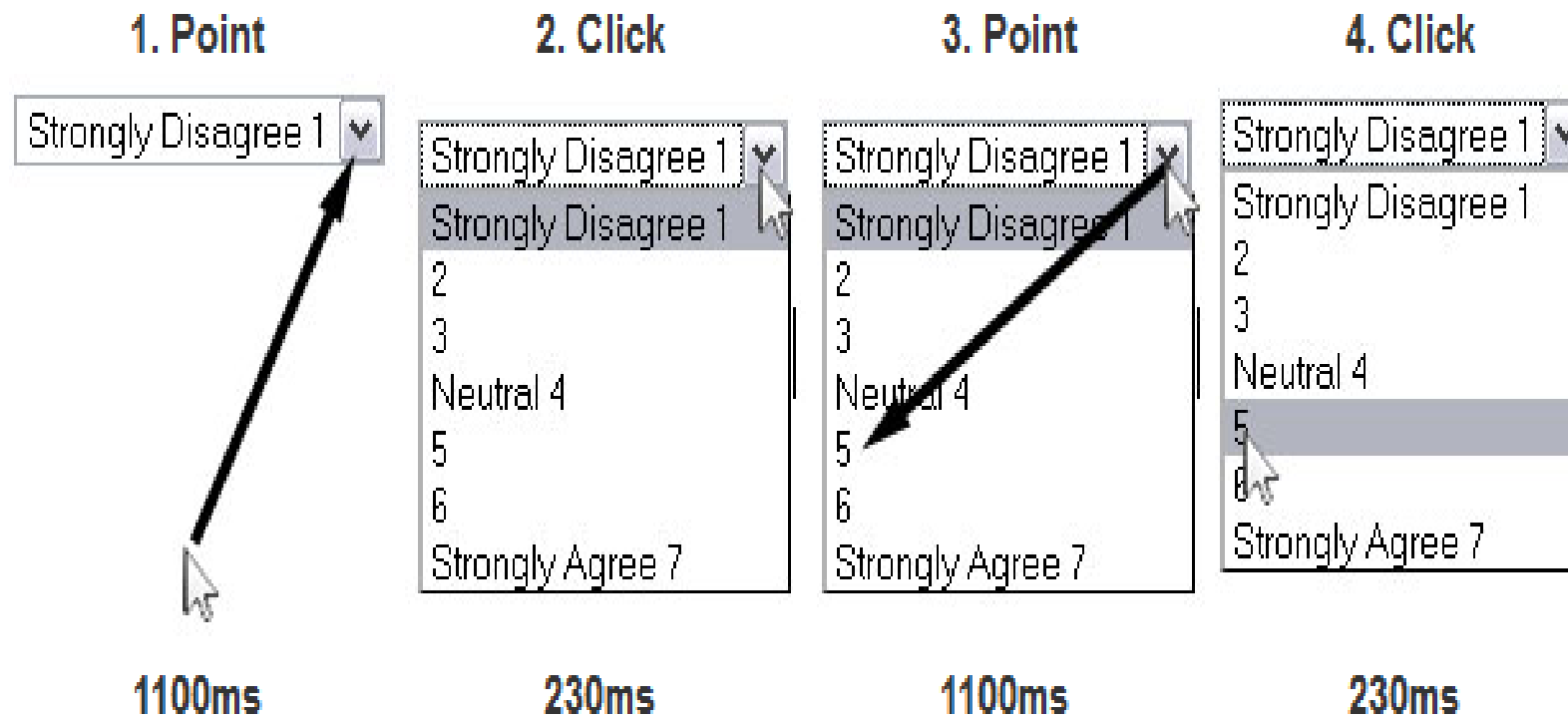


Figure 1: Selecting an option from a drop-down list will take an experienced user 2660ms (2.66 seconds).

Advantages of Predicted Evaluation

- Don't have to build a prototype ,
 - can compare design alternatives with no implementation whatsoever
- Don't have to test real users
- Theory provides explanations of UI problems
 - So it points to the area where design can improve
 - User testing may only reveal problems, not explanations

Limitations of KLM

- Only Expert users doing routine (well-learned) tasks
- Only measures efficiency
 - Not learnability, memorability, errors
- Ignores
 - Errors (Methods must be errors free)
 - Parallel action (shift-click)
 - Planned & problem solving (how does user select method?)
 - Fatigue